S3.1 The periodic table: Classification of elements

Periodicity of ionisation energy, electron affinity and electronegativity

Learning outcomes

By the end of this section you should be able to:

- State the definition for the terms ionisation energy, electron affinity and electronegativity.
- Explain the periodicity of ionisation energy, electron affinity and electronegativity.

When we studied ionic bonding in section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) we looked at how metals have a tendency to lose electrons to form positive ions and non-metals have a tendency to gain electrons to form negative ions. But how might we quantify these tendencies? How might we compare the tendencies to lose or gain electrons between atoms?

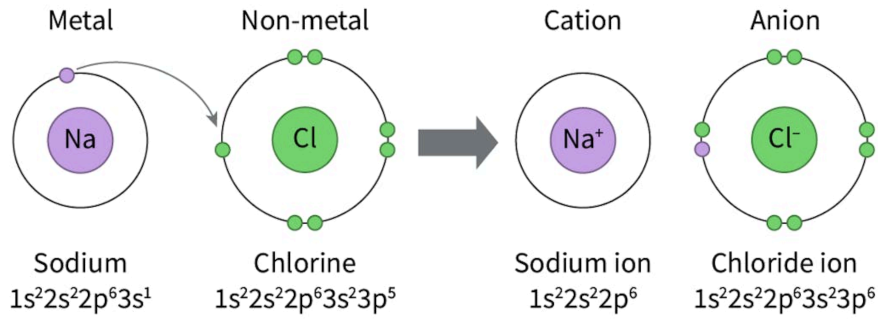

Figure 1. Metals have tendencies to lose electrons and non-metals have tendencies to gain electrons.

More information for figure 1

In **Figure 1** you can see the formation of the sodium cation. You will notice that the valence electron from the n = 3 energy level has been removed to form this cation. How is this electron removed? What is needed for the removal of an electron from an atom?

You may recall from section S1.3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/) that when an electron absorbs enough energy to be promoted to the $n = \infty$ energy level, this represents the point when the electron has been removed from the atom. This is when enough energy has been applied to overcome the attraction between the nucleus and that electron. The atom is said to be ionised once the removal of an electron occurs.

Ionisation energy

Ionisation energy is the energy required to remove the outermost electron from a species. A more specific form of this term is the first ionisation energy. This is the minimum energy required to lose one mole of electrons from one mole of gaseous atoms in the ground state. The qualifiers 'one mole' and 'gaseous' in this definition are used to keep variables constant to directly compare between different elements. The general equation for the first ionisation energy of an element is as follows:

$$X(g) \rightarrow X^+(g) + e^-$$

Using sodium, Na, as an example, the equation is:

$$Na(g) \rightarrow Na^+(g) + e^-$$

Data for the first ionisation energy for all elements can be found in section 9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet. The first ionisation energies are plotted for the first 20 elements in **Figure 2**. Using this graph, identify the elements from the same group (e.g. group 1 H (Z = 1), Li (Z = 3), Na (Z = 11), K (Z = 19)). How would you describe the trend in first ionisation energy going down a group? How would you describe the trend in first ionisation energy radius going across a period? Can you think of explanations for these trends?

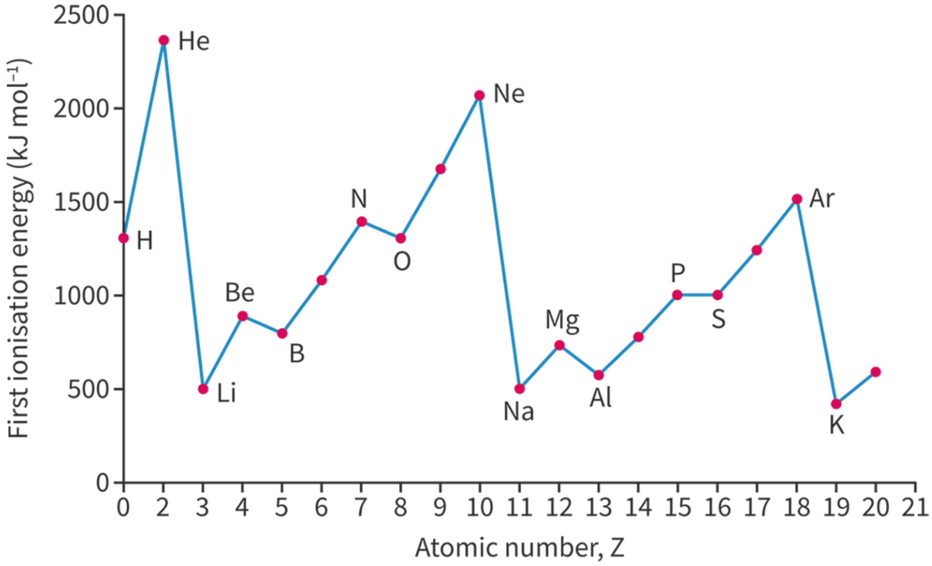

Figure 2. First ionisation energy for elements for the first 20 elements.

More information for figure 2

Table 1 shows the first ionisation energies for elements in groups 1, 2, 17 and 18. Looking at the atomic radii illustrated in **Figure 3**, how does the trend of first ionisation energies compare with that of atomic radii? What do you notice about the trend in first ionisation energy going down a group?

Table 1. First ionisation energy for elements in groups 1, 2,

Period	Highest occupied energy level	Group 1 First ionisation energy (kJ mol^{-1})		Group 2 First ionisation energy (kJ mol^{-1})		Group Firs ionisat ener (kJ mo	
1	$n = 1$	H	1312				
2	$n = 2$	Li	520	Be	900	F	
3	$n = 3$	Na	496	Mg	738	Cl	
4	$n = 4$	K	419	Ca	590	Br	
5	$n = 5$	Rb	403	Sr	549	I	
6	$n = 6$	Cs	376	Ba	503	At	

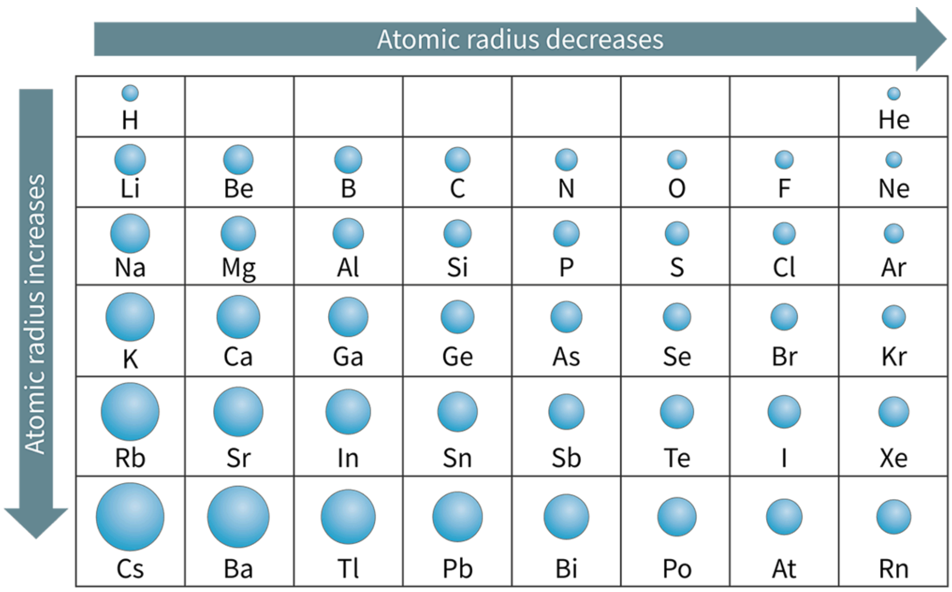

Figure 3. Atomic radii of the elements.

More information for figure 3

Table 2 shows the number of protons, atomic radius and first ionisation energy for period 3 elements. Why would the first ionisation energy increase across the period when valence electrons are all occupying the same *n* = 3 energy level?

Table 2. Number of protons, atomic radius and first ionisation energy

Element	Na	Mg	Al	Si	P
# of protons	11	12	13	14	15
Atomic radius ($\times 10^{-12}$ m)	160	140	124	114	109
First ionisation energy (kJ mol^{-1})	496	738	578	787	1012

General trend in first ionisation down a group

By analysing **Table 1** and **Figure 3**, you can compile the following general trends in first ionisation energy going down a group:

- Going down a group the first ionisation energy decreases.
- Going down a group atomic radii increases and electrons occupy higher energy levels.
- Electrons in higher energy levels have a weaker attraction to the nucleus.
- Less energy is required to remove an electron.

General trend in first ionisation energy across a period

By analysing **Table 2**, you can compile the following general trends in first ionisation energy going across a period:

- Going across a period the first ionisation energy increases.
- Going across a period the nuclear charge increases (there are more protons in the nucleus).
- Going across a period the atomic radius decreases (atoms get smaller except for Ar which increases slightly compared to Cl).

- More energy is required to remove an electron.

So, to summarise, the trend in ionisation energy across a period depends on the nuclear charge and the atomic radius. The greater the nuclear charge and the smaller the atomic radius, the stronger the attraction between the nucleus and the valence electrons and the higher the ionisation energy. The trend in ionisation energy going down a group depends on the atomic radius and the number of occupied energy levels. The more occupied energy levels, the greater the atomic radius which results in a weaker attraction between the nucleus and the valence electrons. This causes the ionisation energy to decrease down a group.

Making connections

You may have noticed here that we are only evaluating general trends (i.e. the general increase or decrease across a period). There are discontinuities within these trends that we will look at more closely in section S1.3.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/) and section S3.1.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/discontinuities-in-ionisation-energy-hl-id-44697/).

Test your knowledge of the trends in ionisation energy by placing a > or < sign in the boxes below to indicate which element has the greater first ionisation energy. For example, Mg > Ca would be used to indicate that Mg has a greater first ionisation energy than Ca.

Interactive 1. Identification of which element has the greater first ionisation energy.

More information for interactive 1

Electron affinity

The term ionisation energy is specific to the absorption of energy to remove an electron to form positive cations. The quantitative term that is specific to the gain of electrons to form negative ions is called electron affinity. If you think about the term affinity, what do you think it might mean? A non-scientific understanding of the word affinity would be to have a natural liking for. You can think of electron affinity in a similar manner, elements with a large absolute value for electron affinity are those with the greatest tendency to gain electrons and those with the lowest absolute value for electron affinity are those with the lowest tendency to gain electrons.

Electron affinity is the energy change that occurs when an electron is added to a species. When a neutral atom gains an electron, energy is released. This is called the first electron affinity and is illustrated in **Figure 4** for hydrogen. Energy is released due to the small force of attraction between the nucleus and the incoming electron.

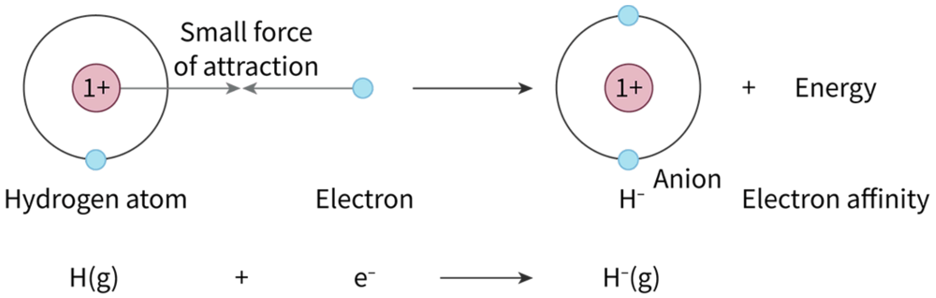

Figure 4. Hydrogen releasing energy as it gains an electron.

More information for figure 4

Section 9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet has values for the first electron affinity of most elements. You will notice that these values are negative, indicating the release of energy. **Table 3** shows the first electron affinity for group 17. How would you describe the general trend of electron affinity down a group?

Table 3. First electron affinity values for group 17 elements.

Group 17	
Element	**Electron affinity (kJ mol^{-1})**
F	—328
Cl	—349
Br	—325
I	—295
At	—270

General trend in electron affinity going down a group

- The absolute value for first electron affinity generally decreases going down a group (lower affinity for gaining an electron).
- Elements further down have a larger atomic radius, with outer valence shells further away from the nucleus.
- Weaker attraction between the added electron and the nucleus.

Table 4 shows the first electron affinity for period 3 elements. How would you describe the general trend of electron affinity across a period? Why do you think Mg and Ar have no values for electron affinity?

Table 4. Electron affinity values for period 3 elements.

Element	Na	Mg	Al	Si	P	S	Cl
Electron affinity (kJ mol^{-1})	—53		—42	—134	—72	—200	—329

General trend in electron affinity going across a period

- Absolute value for first electron affinity generally increases going across a period (higher affinity for gaining an electron).

- Elements across the period have a higher effective nuclear charge.
- Stronger attraction between the added electron and the nucleus.

You may have noticed that there are no values for first electron affinity listed for Mg and Ar. This is because many elements with completely filled subshells do not tend to gain electrons, therefore they do not have an associated electron affinity value.

Test your knowledge of the trends in electron affinity by placing a > or < sign in the boxes below to indicate which element has the greater first electron affinity.

Identify which element has the greater first electron affinity.

Use **<** or **>** to fill in the blanks.

First electron affinity

S ☐ Na

O ☐ F

Li ☐ F

Na ☐ K

Cl ☐ Br

Check

Interactive 2. Identification of which element has the greater first electron affinity.

More information for interactive 2

Electronegativity

Recall from section S2.2.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/) that electronegativity is the tendency of an atom to attract a shared pair of electrons in a covalent bond. Each atom has an assigned electronegativity value that is relative to fluorine. They can be found in section 9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet. Of the quantities discussed in this section, which quantity would you expect the trends in electronegativity to be most similar to? What do you think the general trend for electronegativity would be going down a group? What about going across a period?

Table 5 shows the electronegativity values for group 17. How would you describe the general trend of electronegativity going down a group?

Table 5. Electronegativity values for group 17 elements.

Group 17	
Element	**Electronegativity**
F	4.0
Cl	3.2
Br	3.0
I	2.7
At	2.2

General trend in electronegativity going down a group

- Electronegativity values generally decrease down a group
- Atomic radius increases down a group from additional energy levels
- Weaker attraction between the nucleus and the shared pair of electrons

Table 6 shows the electronegativity values for period 2 elements. How would you describe the general trend across a period? Why do you think the noble gases do not have an electronegativity value assigned to them?

Table 6. Electronegativity values for period 2 eleme

Element	Li	Be	B	C	N
Electronegativity	1.0	1.6	2.0	2.6	3.0

General trend in electronegativity going across a period

- Electronegativity values generally increase going across a period.
- Effective nuclear charge increases going across a period.
- Stronger attraction between the nucleus and the shared pair of electrons.

Aspect: Evidence

Electronegativity values are relative and are a ‘made up’ quantity. These values are calculated by considering the energy it takes to break the bonds between two elements. We assign a lot of value to these numbers (think back to the bonding triangle) but the fact that they are not measurable quantities certainly has its limitations. When scientists encounter evidence that does not fit a model, it calls for both reexamination of the evidence and reexamination of the model. In April 2021, scientists proposed an updated scale for electronegativities in this (https://www.nature.com/articles/s41467-021-22429-0) *Nature Communications* paper.

The noble gases do not have an assigned value for electronegativity as they do not form shared pairs of electrons with other atoms.

Test your knowledge of the trends in electronegativity by placing a > or < sign in the boxes below to indicate which element has the greater electronegativity.

Interactive 3. Identification of which element has the greater electronegativity.

More information for interactive 3

Periodicity of atomic radius, ionisation energy, electron affinity and electronegativity

In both this and the previous section 3.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/), we have been looking at general trends in the periodic table for atomic radius, ionisation energy, electron affinity and electronegativity. **Figure 5** is a periodic table summarising these general trends.

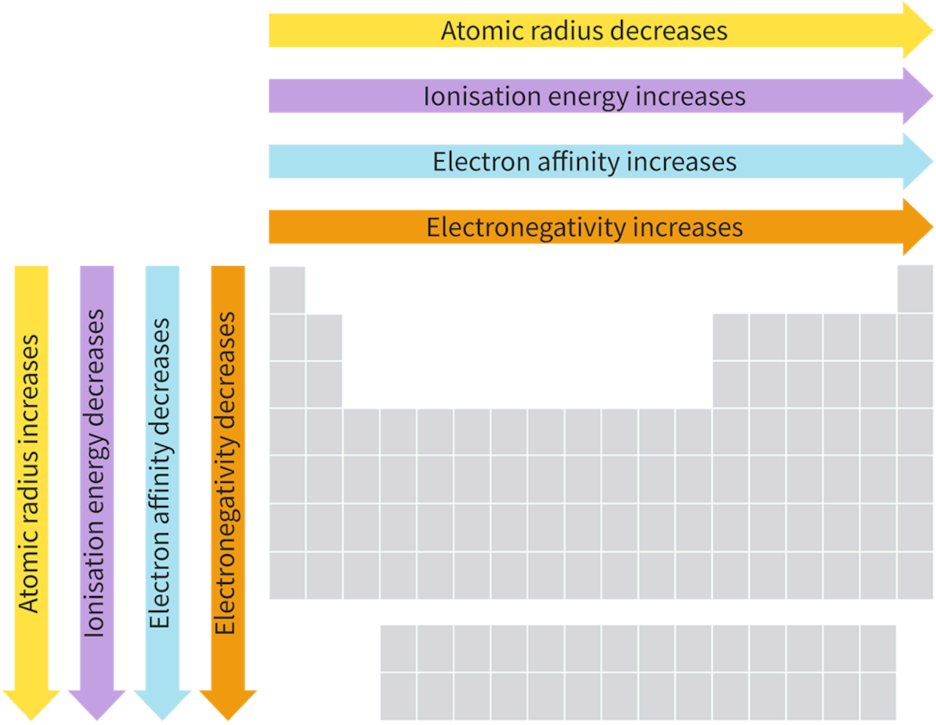

Figure 5. General trends for atomic radius, ionisation energy, electron affinity and electronegativity in the periodic table.

More information for figure 5

Study skills

Periodicity refers to the general trends we observe in the periodic table for different properties. While it is helpful to describe the trends in terms of increasing/decreasing from left to right and top to bottom it is important to note that these are not explanations. Explanations for these trends involve unpacking the attractive forces between the nucleus and the outer valence shell, along with the repulsion between electrons. The strength of the attractive forces is always influenced by the number of occupied energy levels and the effective nuclear charge of an atom.

Activity

- **IB learner profile attribute:** Balanced
- **Approaches to learning:** Research skills — Evaluating information sources for accuracy
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** Group activity

Pasta periodic table

Materials needed (for each student):

- Dry spaghetti (alternatively can use bamboo skewers or pipe cleaners)
- Ruler
- Glue
- Printed copy of the periodic table
- Two pieces of cardboard the same size as your periodic table
- Thumb tack (to poke holes into paper and cardboard)

As a group of four, each group member will select one of the following periodic trends:

- Atomic radius
- Ionisation energy
- Electron affinity
- Electronegativity

You will create a three-dimensional bar graph for your trend using spaghetti to represent the bars. You will need to scale the length of each spaghetti stick to represent value of your chosen parameter. Below provides steps to complete this task.

Three-dimensional bar graph

- Glue both pieces of cardboard together to create one thick piece.
- Glue the periodic table on the top side.
- Using the thumb tack poke a hole through the periodic table and the first cardboard layer for each element.
- Measure the length of your spaghetti stick.
- Decide what measurement you would like each unit to represent. For example, if a spaghetti stick measured 25 cm, then your largest value can only be 25 cm. Let's say the largest value you are representing is 9567. This would be represented by the full spaghetti stick. Now if you wanted to represent 8653 using spaghetti sticks you would simply multiply 25 by $\frac{8653}{9567}$.
- Measure and cut your spaghetti to scale for each value using the following formula:

$$\text{length of full spaghetti} \times \frac{\text{value you are representing}}{\text{largest value from your data set}}$$

- Insert each stick of spaghetti into the appropriate element punched hole in your periodic table.

Once you are finished with your periodic table, help your group members with their spaghetti sticks until all group members are finished. Place all periodic tables side by side.

As a group discuss and complete the following:

- Identify any periodic trends that are evident.
- Discuss the ease to which you were able to identify trends. Were any properties particularly difficult to identify a clear trend? Share why you think these are difficult.
- In ChatGPT ask the AI chatbot to explain each trend down a group and across a period. The AI chatbot will likely produce a reasonable but incomplete response. As a group evaluate the response. What parts of the

response were well crafted? What parts of the response are missing for a high-level chemistry explanation?